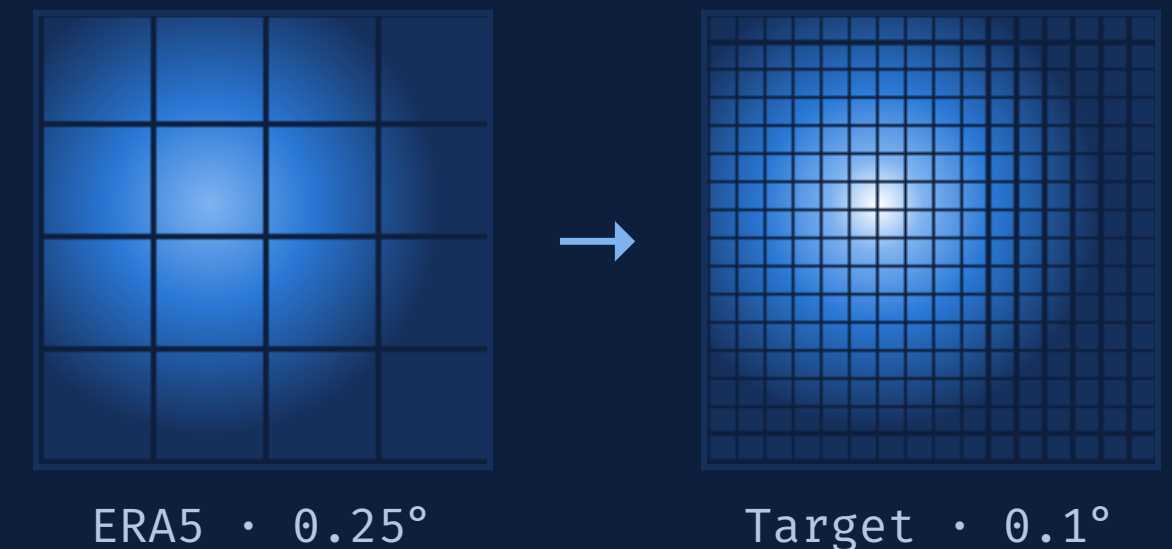


Physics-Constrained Rectified Flow for Climate Downscaling

A projection at inference makes rain non-negative and matches the coarse mean for any model. At a matched three-year budget, the training penalties add no measurable quality.



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The talk in one slide

A **projection at inference** makes rain non-negative and matches the coarse mean for any model. At a matched three-year budget, the **training penalties add no measurable quality**.

45%

of CorrDiff (ours)
ensemble-mean rain
cells below zero on
ERA5 2020, at the best
PSNR in the set

0

NVR and MCE after
the projection, for any
model, up to float32
rounding

RF-base

PC-RF

0.942 0.939

SSIM, RF-base vs PC-RF, matched three-
year budget

ROADMAP

01	Problem	3–5
02	Method	6–9
03	Evidence	10–13
04	Checks	14–16
05	Limits	17–19

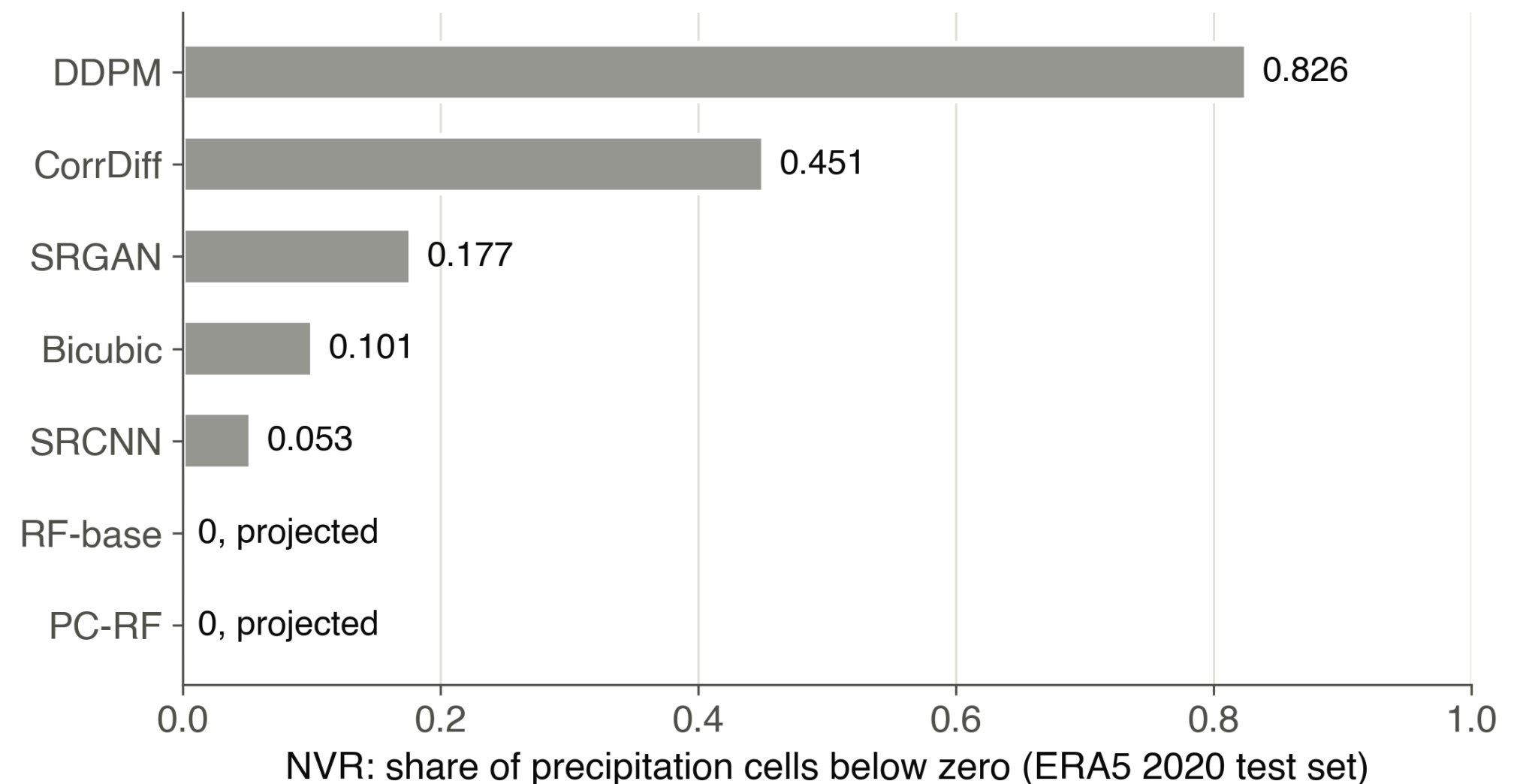
Sharp fields can rain below zero

NVR, CORRDIFF (OURS), ERA5 2020

45%

of ensemble-mean rain cells below zero, at the best PSNR in the set, 28.38

- Bicubic fails too, at 0.101, from interpolation overshoot at rain edges.
- On 2019–2021, CorrDiff (ours) falls to 0.108, near bicubic at 0.119.
- Negative rain has no physical meaning in a water balance.
- Only the two projected models sit exactly at zero.



Data from Table 2 of the paper. SRCNN, SRGAN, DDPM and CorrDiff are our reimplementations, scored without the projection, which any of them could apply.

Two kinds of evidence

SCORES

SSIM $\uparrow$ · PSNR $\uparrow$ · CRPS $\downarrow$

- Need a high-resolution reference.
- Inherit every error in that reference.
- Rank models, and mix rain with two wind channels here.
- Our reference is interpolated ERA5, so they cannot show downscaling skill.

DIAGNOSTIC

DE

- Divergence of the wind field.
- Carries no guarantee, because wind is not projected.
- On ERA5, a sign error makes it a deformation term.

CERTIFICATES

NVR $\downarrow$ · MCE $\downarrow$

- Need only the output and the coarse input.
- Anyone can run them in one cheap pass.
- Zero after the projection, for any model.
- Certify the serving step, so they cannot rank two projected models.

AIAS⁺ 2026 CALL

Reliability, robustness, safety, and human oversight in scientific AI systems

Benchmarks, evaluation methods, and infrastructure for AI-accelerated science

Guarantees go in the inference path

Use the training loss to **shape** the field. Use a projection at inference to **certify** it.

- Three differentiable penalties act during training: divergence, negative rain, and drift of the domain mean.
- One projection after sampling makes two constraints hold for any model: rain is non-negative, and the domain mean matches the coarse input.
- Both properties are checkable from the output and the coarse input alone, with no high-resolution reference.
- Wind is not projected, so divergence stays a diagnostic.

TRAINING

Flow loss + three penalties shape the field

SAMPLING

8 members $\times$ 20 Euler steps draw the field

PROJECTION

Clamp, then one rescale: $\text{NVR} = \text{MCE} = 0$

CHECKS

Reference-free gate, then scores

Data and setup

COARSE INPUT

ERA5 precipitation and 10 m u, v at 0.25°
40–50° N, 10° W–5° E, at 00, 06, 12 and 18
UTC
Years 2019–2021

FINE TARGET

ERA5-Land precipitation at 0.1° , about 8 to
11 km, resampled to 128×128
Ocean cells set to zero, since the product covers
land only
No fine wind target exists, so the wind target is
the upsampled coarse wind

TRAINING AND BASELINES

Chronological 80/10/10 splits on 2020 (1,171
training samples) and on 2019–2021. The
2020 test set runs late November to 31
December
Main run: 300 epochs, batch 32, AdamW at
 2×10^{-4} , one A100, about 6.2 GPU-hours, one
seed per arm
Baselines: bicubic and our SRCNN, SRGAN,
DDPM and CorrDiff, scored without the
projection

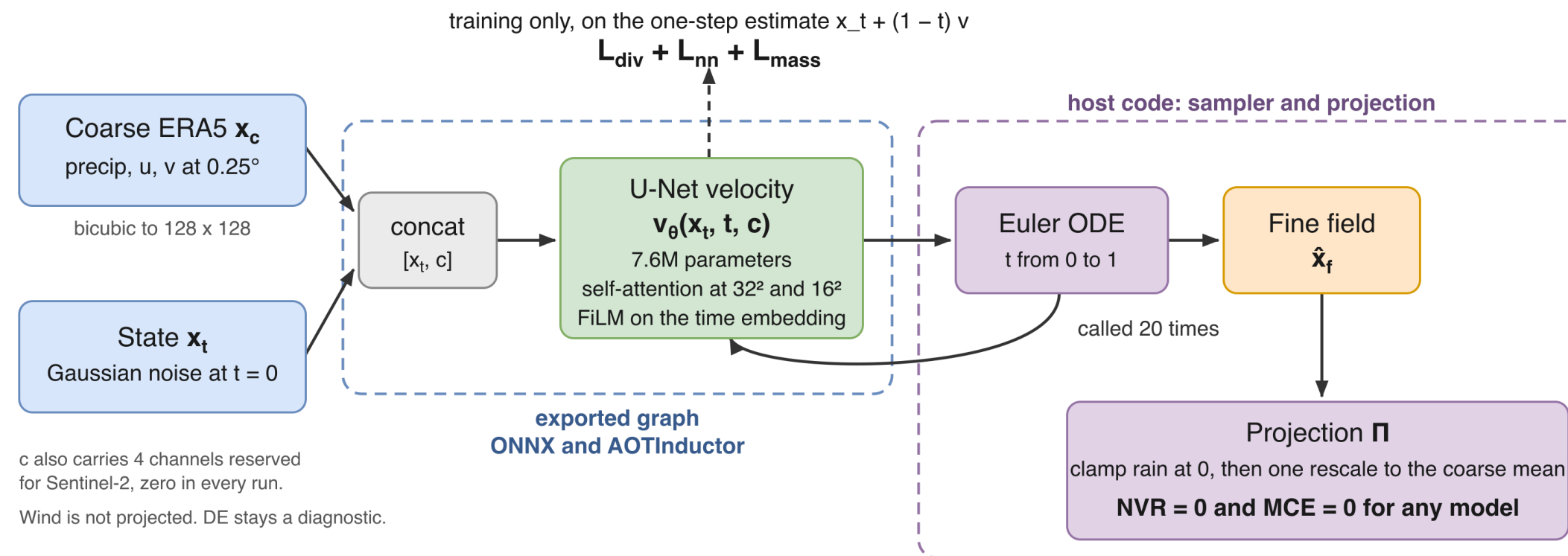
REFERENCE CAVEAT

ERA5-Land precipitation is interpolated from ERA5 forcing over a different accumulation
window. The perceptual scores therefore measure agreement with an interpolant and
cannot establish downscaling skill. The validity checks never use the reference.

CORRECTIONS TO THE ABSTRACT

The target grid is about 8 to 11 km.
The Sentinel-2 channels were zero in every
run.

A 7.6M-parameter conditional U-Net



Adapted from Figure 1 of the paper. PC-RF as trained and served. The dashed blue region is the only exported graph. The purple region is host code.

RECTIFIED FLOW

$$L_{RF} = \mathbb{E} \| v_\theta(x_p, t, c) - (x_1 - x_0) \|^2$$

$$\hat{x}_1 = x_t + (1 - t) v_\theta(x_p, t, c)$$

The velocity runs from Gaussian noise to the data, so sampling is a short ODE solve. The penalties act on the one-step clean estimate $\hat{x}_1$.

- Coarse conditioning is concatenated with the state at the input, and FiLM carries the time embedding.
- Inference: 8 ensemble members, 20 Euler steps each, then the projection.
- The penalties never appear at inference.

Three penalties shape the field in training

$$L = L_{\text{RF}} + \lambda_1 L_{\text{div}} + \lambda_2 L_{\text{nn}} + \lambda_3 L_{\text{mass}}$$

with $\lambda_1 = \lambda_2 = \lambda_3 = 1$

L_{div} Squared divergence of (u, v) on the clean estimate, by central differences

L_{nn} Squared negative part of the rain channel

L_{mass} Squared drift of the domain mean from the coarse mean

KNOWN ERROR ON ERA5

Grid rows run north to south, so a latitude sign error makes the operator compute a deformation term, not the divergence. Every ERA5 run trained with L_{div} on that term.

```
x1_hat = x_t + (1 - t) * v      # one-step clean estimate
div     = ddx(u) + ddy(v_wind)  # central differences
L_div   = (div ** 2).mean()
L_nn    = (p.clamp(max=0) ** 2).mean()
L_mass  = ((p.mean((1,2,3)) - p_coarse_mean) ** 2).mean()
loss    = L_rf + L_div + L_nn + L_mass
```

Simplified from src/pcrf/physics.py and flow.py. Autograd handles the rest, at a few elementwise ops per step.

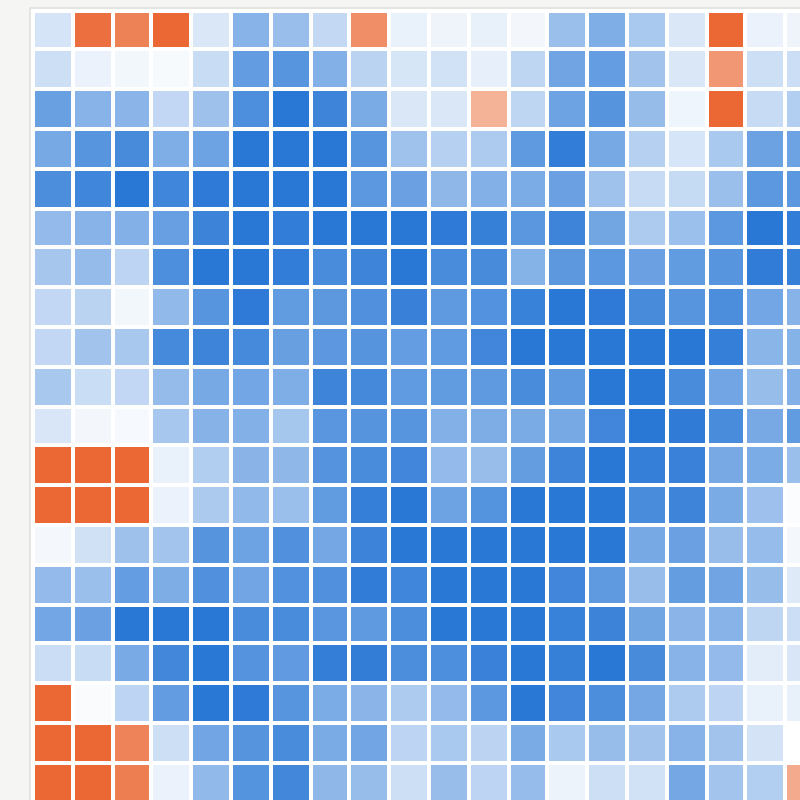
The projection is two lines of host code

$$\Pi(p) = \max(p, 0) \cdot \frac{\bar{p}_c}{\text{mean}(\max(p, 0))}$$

```
p = x[:, 0:1].clamp(min=0.0)
p = p * (coarse_mean / p.mean((1,2,3), keepdim=True))
```

- The clamp removes negative rain, which raises the mean. One factor per sample then sets the mean equal to the coarse mean $\bar{p}_c$.
- A single factor keeps every ratio between cells, so the pattern stays and only the amplitude moves.
- NVR and MCE drop to zero for any model, up to float32 rounding. The coarse mean covers land and sea.
- Wind is not projected. A divergence-free projection needs a Poisson solve, and near-surface convergence is physically real.

LIVE ILLUSTRATION • SYNTHETIC FIELD, NOT ERA5



NVR

0.055

mean(p)

0.548

MCE

0.0000

coarse

0.548

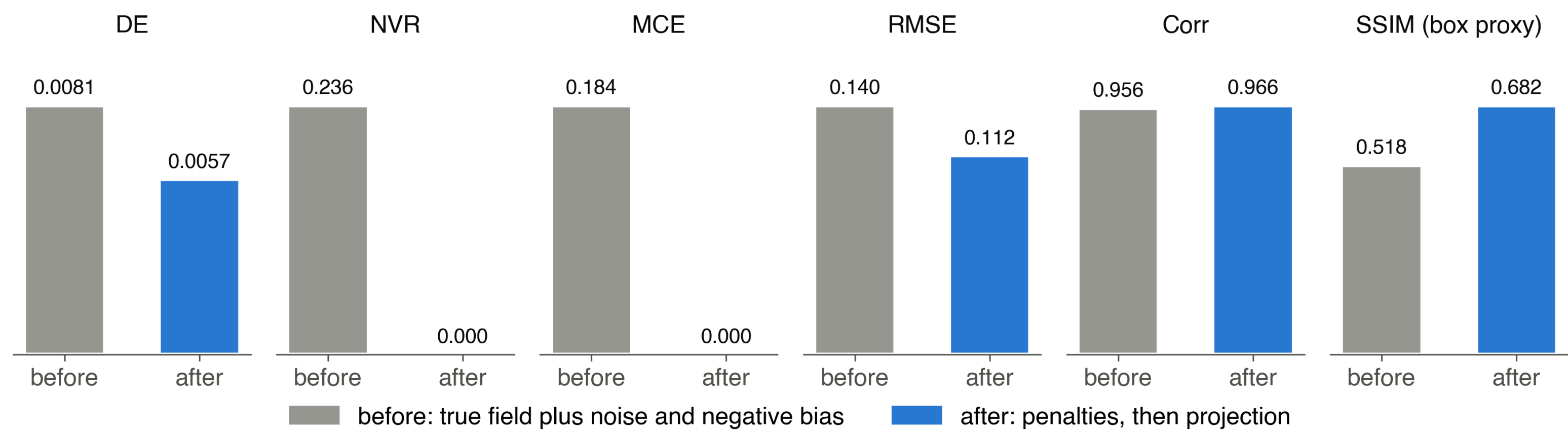
Stage 1 of 3 shows a synthetic raw field. Orange cells are negative rain.

1 • Raw field

2 • Clamp at zero

3 • Rescale the mean

Synthetic check: the mechanism works



- One constructed 128×128 field with exact physics, plus correlated noise and a negative bias. No network is trained.
- The projection drives NVR and MCE to zero. The penalties cut divergence error by about 30%, which the projection cannot touch.
- Both steps run together, so the RMSE and SSIM gain cannot be split between them. Grid coordinates, so the ERA5 sign error does not apply.

NVR, BEFORE AND AFTER

0.236 → 0.000

ERA5 2020: only projected models are exactly valid

Model	SSIM ↑	PSNR ↑	CRPS ↓	DE	NVR ↓	MCE ↓
Bicubic	0.885	26.19	1.356	0.0648	0.101	0.0003
SRCNN (ours)	0.833	27.74	1.461	0.0738	0.053	11.14
SRGAN (ours)	0.855	27.03	1.506	0.0700	0.177	13.22
DDPM (ours)	0.656	26.18	1.609	0.0715	0.826	7.50
CorrDiff (ours)	0.864	28.38	1.092	0.0679	0.451	6.48
■ RF-base, 200 epochs	0.912	26.39	1.188	0.0643	0	0
■ PC-RF, 300 epochs	0.914	26.59	1.149	0.0644	0	0

Table 2 of the paper. Bold marks the best value per column. CRPS uses the 8 members, and the other metrics use their mean. MCE is a relative error of the domain mean, not a water amount. On ERA5, a latitude sign error makes DE a deformation term, so its column has no arrow and no bold.

HOW TO READ IT

- Every unprojected model fails NVR, bicubic included. The two projected rows are the only exactly valid ones.
- RF-base and PC-RF had different budgets, 200 versus 300 epochs, so that pair is unmatched.
- SSIM, PSNR and CRPS average rain with two wind channels whose target is upsampled coarse wind.
- CorrDiff (ours) was never tested with our projection, which it could apply.

Matched one-year budget: tied on SSIM

Arm, 200 epochs, batch 16	SSIM ↑	PSNR ↑	CRPS ↓	DE
■ Full PC-RF	0.912	26.54	1.161	0.0634
No divergence penalty	0.912	26.58	1.146	0.0649
No non-negativity penalty	0.913	26.55	1.160	0.0644
No mass penalty	0.915	26.50	1.166	0.0639
■ RF-base, no penalties	0.913	26.40	1.186	0.0643

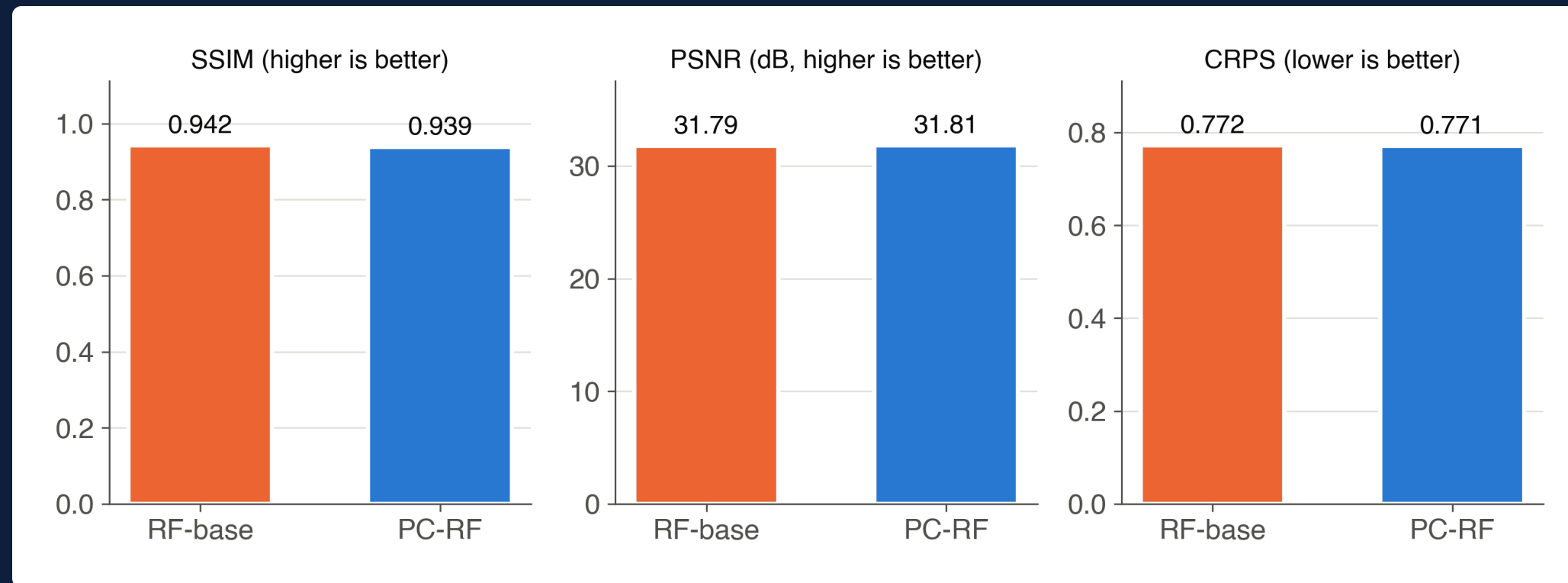
Table 3 of the paper, one-year block. All arms are projected, so NVR and MCE are zero throughout. On ERA5, the divergence penalty and DE act on a deformation term. The RF-base checkpoint is the Table 2 checkpoint, rescored with a fresh 8-member sample.

SSIM SPREAD ACROSS ALL FIVE ARMS

0.003

- Every penalty arm ties RF-base on SSIM, between 0.912 and 0.915.
- The penalty arms lead slightly on PSNR and CRPS. With one seed per arm, that lead cannot be separated from run-to-run variation.
- Dropping any single penalty changes nothing measurable.

Three years, matched budget: still a tie



Data from Table 4 of the paper. Every arm trained 100 epochs on 2019–2021. Orange is RF-base, blue is PC-RF. These scores mix rain with two wind channels.

■ RF-base SSIM

■ PC-RF SSIM

0.942 0.939

- PSNR and CRPS show the same tie. Both models keep NVR and MCE at zero, because both are projected.
- The penalties were cheap, and on this data they bought no measurable quality. The projection did all of the certifying.
- The tie fits the thesis, and a reference that breaks the mass relation could also explain it.

TWO CAVEATS

The scores measure agreement with an interpolated reference, and each arm has one seed. A tie shows what we measured, not that the penalties can never help.

What the checks certify

Reference-free

NVR and MCE need only the generated field and its coarse input. Reanalysis references carry their own errors, and these two never touch one.

Certificates

After the projection both are zero for any model, up to float32 rounding. They report the serving step, not the weights, so they cannot rank two projected models.

Diagnostic

DE carries no guarantee. On ERA5, a latitude sign error makes it a deformation term, and every DE value on this data reflects that term.

Necessary, not sufficient

A field can pass every check and still be wrong.

REFERENCE WIND, DEFORMATION TERM

0.0648 0.0567

ERA5 2020

2019–2021

The reference wind is itself upsampled coarse wind. In both main tables, PC-RF and RF-base sit within 1% of it on the same term.

A protocol for scoring downscalers

STEP 1 • GATE

$NVR = 0, MCE = 0$

Reference-free, a few tensor ops,
needs only the output and the
coarse input.

A field that fails is not scored further.

Certifies the serving step



STEP 2 • DIAGNOSE

DE beside the gate

No guarantee, because wind is
not projected.

Know what your operator computes.
Ours computed a deformation term
on ERA5.

Flags, does not certify



STEP 3 • SCORE

SSIM, PSNR, CRPS

Compare to the reference, and say
what that reference can support.

Match training budgets and use
several seeds before reading a ranking.

Ranks, given a good reference

On our data, **Step 1 is the only part of this protocol that certifies anything.**

Export the network, not the sampler

EXPORTED GRAPH

$$v_{\theta}(x, t, c)$$

One velocity call. Small, stateless, serves any step count or solver.

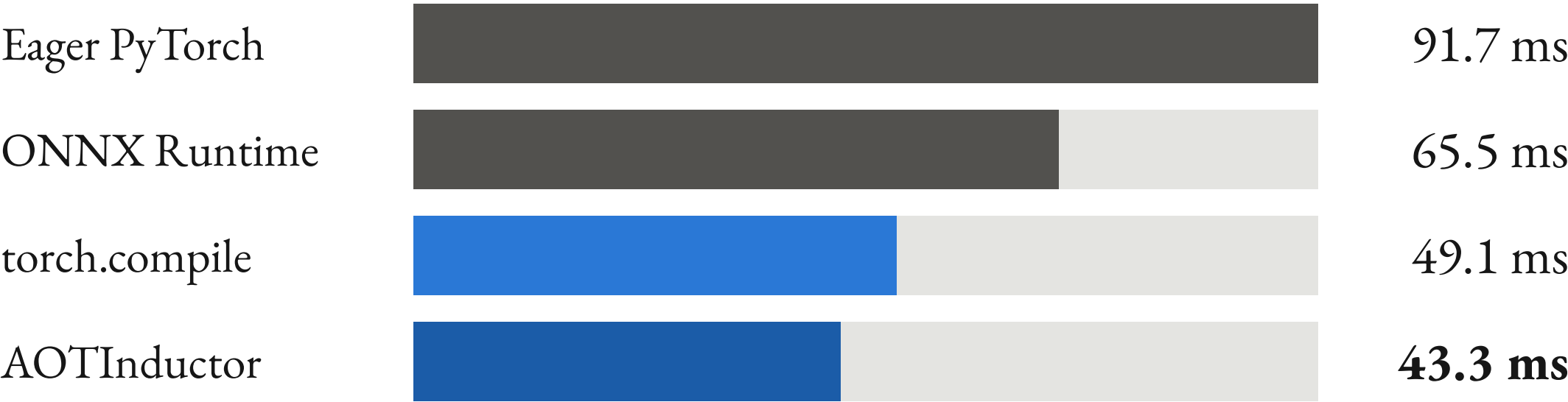
HOST CODE, PYTHON AND RUST

Euler loop: 20 calls and an add

Projection: clamp, then one rescale

The cost is keeping the copies equal. Today each language tests its own.

ONE REQUEST, P50, 20 STEPS + PROJECTION, BATCH 1, A100-40GB



- In a separate run with TF32 off, torch.compile and AOTInductor match eager within 2×10^{-6} . NVR stays at zero on every backend.
- ONNX Runtime copies through the host every step.
- Caveats: random weights in the trained architecture, and the Rust service and Ray Serve ran on a demo-scale export.

Limitations we raise ourselves

- 1 The reference cannot show downscaling skill. It is ERA5 interpolated over a different accumulation window, so the perceptual tables rank agreement with an interpolant.
- 2 The projection matches the coarse mean over land and sea, a relation the reference breaks by zeroing the sea.
- 3 One domain, one test season, one seed per arm, and no extremes evaluation.
- 4 Baselines were scored without the projection, which any of them could apply.
- 5 No independent fine target has tested the mass penalty yet.

KNOWN ERROR ON ERA5

A latitude sign error made the divergence penalty and DE act on a deformation term, not divergence.

The synthetic check, in grid coordinates, is unaffected. Every ERA5 number that depends on L_{div} or DE carries this caveat.

Next steps

INDEPENDENT TARGET

A regional reanalysis with its own precipitation, over matched accumulation windows, could measure downscaling skill and test the mass penalty for the first time.

PROJECT THE BASELINES

Score every baseline with the projection. It is cheap, and it would show whether our CorrDiff keeps its PSNR and CRPS lead once it passes the gate.

SEEDS AND EXTREMES

Several seeds per arm, and an evaluation conditioned on extreme events, where negative cells and mass drift matter most.

ONE PROJECTION SPEC

Test the Python and Rust copies of the projection against one shared specification and one shared fixture.

RERUN WITH THE FIX

Correct the latitude sign, penalize divergence relative to the upsampled coarse wind, retrain the penalty arms, and recompute DE.

WIND

Decide whether a divergence projection is wanted at all, given that near-surface convergence is physically real.

1

The guarantee lives in the projection, and it holds for any model.

2

NVR and MCE certify the serving step. They cannot rank two projected models.

3

At a matched three-year budget, the penalties added no measurable quality on this data.

Questions

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Metric definitions

Metric	Definition	Input
NVR ↓	Share of precipitation cells below zero	Ensemble mean, no reference
MCE ↓	Relative error of the domain mean against the coarse mean, not a water amount	Ensemble mean and coarse input
DE	Divergence error in standardized grid units. On ERA5 the operator computes a deformation term	Ensemble-mean wind, no reference
SSIM ↑, PSNR ↑	Structural similarity and peak signal-to-noise ratio, averaged over rain and two wind channels	Ensemble mean and reference
CRPS ↓	Continuous ranked probability score over the 8 members, averaged over rain and two wind channels	All 8 members and reference

Training configuration

NETWORK

Parameters	7.6M, conditional U-Net over one 128×128 grid
Conditioning	Coarse fields concatenated with the state at the input
Time embedding	FiLM
Objective	Rectified flow, plus three penalties at $\lambda = 1$ on the one-step clean estimate

BUDGETS

Main run, Table 2	PC-RF 300 epochs, batch 32, AdamW at 2×10^{-4} , one A100, about 6.2 GPU-hours
RF-base, Table 2	200 epochs, batch 16
One-year ablation	Every arm 200 epochs, batch 16
Three-year run	Every arm 100 epochs on 2019–2021

INFERENCE

Ensemble	8 members
Solver	20 Euler steps per member
Projection	Clamp at zero, then one rescale to the coarse mean over land and sea
Wind	Not projected

DATA

Splits	Chronological 80/10/10 on 2020 (1,171 training samples) and on 2019–2021
Test season	Late November to 31 December 2020
Seeds	One per arm
Sentinel-2	Channels zero in every run

Serving details

BENCHMARK

Hardware	One A100-SXM4-40GB
Request	20 Euler steps plus the projection, batch 1
Eager PyTorch	91.7 ms
AOTInductor	43.3 ms
torch.compile	49.1 ms
ONNX Runtime	65.5 ms, copies through the host every step

CORRECTNESS AND SCOPE

Numerics	With TF32 off, torch.compile and AOTInductor match eager within 2×10^{-6}
Validity	NVR stays at zero on every backend
Weights	Random, in the trained architecture
Rust and Ray Serve	Tested on a demo-scale export
Training hardware	Every model trained on one GPU